SAC, TD3, and the HSiKAN backbone

How a signed-hypergraph policy is applied to off-policy continuous control

Aiko (Claude Code) for Dr. Csaba Hajdu — prepared for Kato-sensei

2026-06-22

Abstract

A short technical note for the grasping collaboration. It states the two off-policy actor–critic algorithms we use — TD3 (deterministic) and SAC (maximum-entropy, stochastic) — and then shows precisely how the *HSiKAN* signed-hypergraph network is dropped in as the shared actor/critic representation that reads the robot's kinematic hypergraph. The methodological point is the clean ablation: **fix the algorithm, swap only the backbone** (HSiKAN vs. a parameter-matched MLP), so any difference is attributable to the structure, not to capacity. For the full DDPG → TD3 → SAC lineage and the newer architectures (REDQ/TQC/DroQ/CrossQ/TD7), see the companion survey `reports/2026-06-21-offpolicy-rl-survey.pdf`; for the HyMeKo "MDP-as-data" substrate see `reports/2026-06-21-rl-on-hymeko-technical-report.pdf`. This note is the application-focused subset.

1 The off-policy contract

All methods here learn a deterministic or stochastic policy π and an action-value critic $Q(s, a)$ from a replay buffer of transitions (s, a, r, s', d), off-policy. Two conventions matter for robot tasks:

- **Twin critics / clipped double-Q.** Two critics Q_1, Q_2; the target uses $\min_i Q_i$ to curb the systematic over-estimation bootstrapping induces.
- **Time-limit truncation is *not* a terminal.** When an episode ends only because it hit the step cap, the Bellman target still bootstraps ($d = 0$); treating truncation as $d = 1$ would teach the critic the world ends at the horizon. (This was a real bug we fixed; both trainers now store truncation as non-terminal.)

2 TD3 — twin-delayed deterministic policy gradient

A deterministic actor $\mu_\theta(s)$ and twin critics. Target actions are smoothed; the actor and target networks update less often than the critics. With target nets $\bar{\mu}, \bar{Q}_i$ and Polyak rate τ:

$$\tilde{a}' = \mathrm{clip}\big(\bar{\mu}(s') + \mathrm{clip}(\epsilon, -c, c), -a_{\max}, a_{\max}\big), \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

$$y = r + \gamma(1 - d) \min_{i=1,2} \bar{Q}_i(s', \tilde{a}'), \tag{2}$$

$$\mathcal{L}_{Q_i} = \big(Q_i(s, a) - y\big)^2, \qquad \nabla_\theta J = \nabla_\theta Q_1\big(s, \mu_\theta(s)\big) \text{ (every } \texttt{policy\_delay} \text{ steps)}. \tag{3}$$

Three patches over DDPG: (1) clipped double-Q, (2) delayed policy/target updates, (3) target-policy smoothing. TD3 is the reliable deterministic baseline. Exploration is injected as Gaussian *action noise* at collection time.

3 SAC — maximum-entropy, stochastic

SAC maximises return *plus* policy entropy, $J = \sum_t \mathbb{E}\big[r_t + \alpha\, \mathcal{H}(\pi(\cdot|s_t))\big]$. A squashed-Gaussian actor $a = a_{\max} \tanh(\mu_\phi(s) + \sigma_\phi(s) \odot \varepsilon)$, $\varepsilon \sim \mathcal{N}(0, I)$, with the tanh change-of-variables correction in $\log\pi$; twin soft-Q critics; and the temperature α auto-tuned to a target entropy $\bar{\mathcal{H}} = -\dim(a)$:

$$y = r + \gamma(1-d)\Big(\min_i \bar{Q}_i(s', a') - \alpha \log \pi_\phi(a'|s')\Big), \quad a' \sim \pi_\phi(\cdot|s'), \tag{4}$$

$$\mathcal{L}_\pi = \mathbb{E}\big[\alpha \log \pi_\phi(a|s) - \min_i Q_i(s,a)\big], \quad a \sim \pi_\phi(\cdot|s) \text{ (reparameterised)}, \tag{5}$$

$$\mathcal{L}_\alpha = -\,\mathbb{E}\big[\log\alpha\,(\log\pi_\phi(a|s) + \bar{\mathcal{H}})\big]. \tag{6}$$

SAC is the single strongest, most robust continuous-control baseline — the fair opponent for the structural policy. **The entropy term $\alpha\,\mathcal{H}(\pi)$ is also the research seat:** it is the explicit insertion point where a HyMeKo-native *structural* entropy of the state hypergraph could later be substituted or added (a separate experiment; flagged here only so the seat is visible).

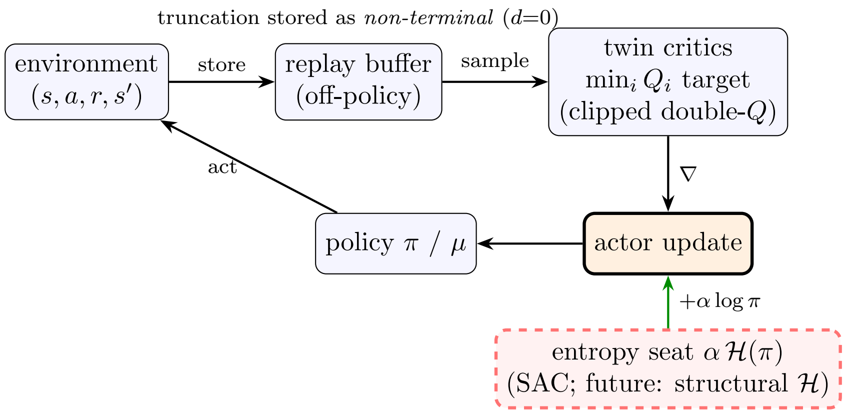

Fig. 1 — the off-policy loop. TD3 and SAC share this skeleton; they differ in the policy (deterministic vs. stochastic) and in whether the orange entropy seat is active. The seat is where a HyMeKo-native structural entropy could later enter.

4 The HSiKAN backbone — reading the kinematic hypergraph

The robot is described once in HyMeKo and compiled to a *signed kinematic hypergraph*: vertices are links, and the signed adjacency (A^+, A^-) encodes the joint connectivity (sign carries the structural relation). `HypergraphState.from_mjcf` derives (A^+, A^-) from the same source that yields the MuJoCo scene, so the policy reads the *compiled structure*, not raw joint indices.

Input. Per-vertex features $x \in \mathbb{R}^{B\times N\times d}$: each link vertex carries its dynamic state (e.g. $[q, \dot{q}]$, plus task channels such as the goal vector or the coin direction).

Layer. One signed graph-conv aggregates self, signed-positive and signed-negative neighbourhoods (a dense, batched SGCN variant — kinematic graphs are tiny and the obs is a rollout batch):

$$h^{(\ell+1)} = \tanh\Big(W_{\text{self}}\,h^{(\ell)} \;+\; W_+\,(A^+h^{(\ell)}) \;+\; W_-\,(A^-h^{(\ell)})\Big), \tag{7}$$

with $A^{\pm}h$ computed as a batched contraction $\sum_m A_{nm}\,h_{bmd}$. After L layers the vertex embeddings are mean-pooled to a graph embedding $z \in \mathbb{R}^{B\times\text{hidden}}$.

Two variants.

- `hsikan` — $A^{\pm}$ are *fixed buffers* (the kinematic structure is given).
- `signedkan` — $A^{\pm}$ are *trainable parameters* initialised from the kinematic adjacency, so the learned weights *are* the weighted star edges of the hypergraph (the fully native "weights-as-hypergraph" form).

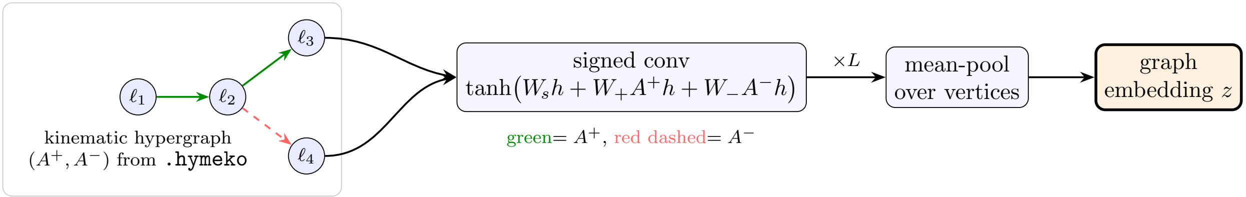

Fig. 2 — HSiKAN signed message passing. Per-vertex dynamic state is aggregated over the signed kinematic adjacency (self / positive / negative), L times, then pooled to a fixed-width embedding z that feeds the actor and critic heads.

5 How it plugs into SAC and TD3

The actor and critic are one shared backbone with two heads (`ActorCritic`). The backbone is the swap point: `build_sac(kind,...)` / `build_offpolicy(kind,...)` take `kind` $\in$ {`hsikan`, `signedkan`, `mlp`}. The HSiKAN actor outputs z to the SAC squashed-Gaussian head $(\mu, \log\sigma)$ or the TD3 deterministic head ($a_{\max}\tanh$); the critic concatenates z with the action. **Nothing in the SAC/TD3 update changes** — only the representation that reads the state.

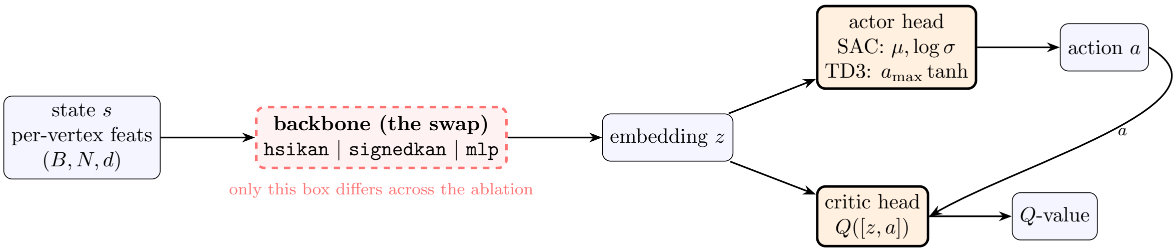

Fig. 3 — fix the algorithm, swap the backbone. The actor and critic share one backbone; HSiKAN (structure-reading) vs. a params-matched MLP (flat) is a single dashed-box swap, so any performance difference is attributable to the representation, not the algorithm or capacity.

role	HSiKAN form	MLP control
state read	signed message passing over (A^+, A^-)	flattened $(N \times d)$ vector
actor head	SAC squashed-Gaussian / TD3 tanh	identical
critic	$Q(\,[z, a]\,)$ on the same backbone family	identical

6 The ablation methodology (what the current run measures)

The question cart-pole could not answer (2 vertices: structure is not load-bearing) is being tested fairly:

- **Fix the algorithm, swap the backbone.** HSiKAN vs. MLP under one shared SAC.
- **Parameter-matched control.** The MLP width is auto-tuned so its parameter count matches HSiKAN's (within $< 1.5\%$) per task — a difference must come from *structure*, not capacity.
- **Real topology, four tasks.** Inverted pendulum (2-vtx, the negative control), Galambos planar coin-grasp (6-vtx, Galambos-sensei's scenario), 6-DOF arm reaching (6-vtx), quadruped goal-reaching (14-vtx, branching) — structure grows across the ladder.
- **Honest metric.** *Curve-max* (best mean-return over the eval schedule), reported as median $\pm$ IQR / worst over 5 seeds. A verdict is credited only when the HSiKAN–MLP median gap *exceeds the cross-seed IQR*; a gap inside the IQR is a tie, never a single-seed claim.

Prior single-seed evidence (cart-pole, and a first Galambos run) showed *parity* with matched capacity, with a whiff of HSiKAN parameter-efficiency only as topology got rich (quadruped). The multi-seed off-policy run now in flight is what turns that into a firm statement; results tables (`booktabs`, drop-in) are generated automatically from the run journal.

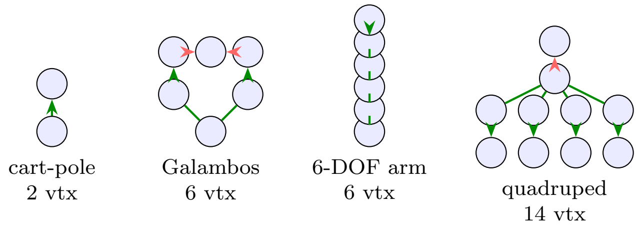

Fig. 4 — the topology ladder. Structure grows left to right; the hypothesis is that any HSiKAN advantage over a flat MLP should also grow with it. Cart-pole is the negative control (structure not load-bearing).

7 Why this matters for the grasping collaboration

The deliverable is a *structurally accountable* control head: the same HyMeKo hypergraph that describes the robot is what the policy reads, the policy network itself round-trips back to HyMeKo, and the comparison against a matched MLP is built in — so a claim that "the structure helps" is always testable, never asserted. If the edge widens on the richer-topology tasks (the hand / contact graphs in grasping are far richer than these testbeds), that is the first real signal that reading the kinematic structure — rather than a flat state — earns its place.